# Feature Selection for Multivariate Time-Series Data After Tensor Reduction

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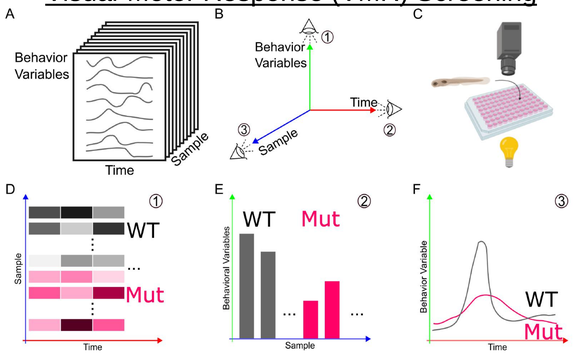
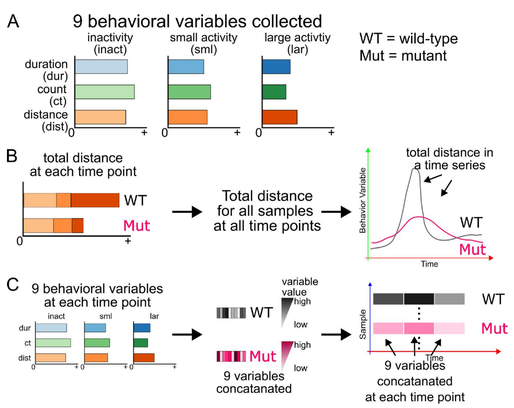
## Background

### Multivariate Time-Series Data From Zebrafish Visual-Motor Response (VMR) Screening




Animals display dynamic behaviors upon stimulation, reflecting the underlying neural processing of the environment. The dynamic behaviors can be captured by behavioral assays, which generate multivariate time-series data known as three-tensor (3T) data. An example is the zebrafish visual-motor response (VMR) data. Proper analysis of 3T data can facilitate the studies on neural processing. Yet, current multivariate analysis (MVA) is incompatible with the 3T data because MVA only handles two-tensor (2T) data structures. One solution is to reduce the 3T data to 2T by engineering features and reduce the dimensions in one tensor.




For instance, VMR assay records many behavioral variables and generates multivariate time-series data. There are two strategies to reduce the tensors: 1) constructing a single behavioral variable, e.g., total distance, and 2) concatenating all behavioral variables at each time point. However, the engineered features may not be optimal for downstream multivariate analysis because 1) key variables may be lost, failing to capture the rich repertoire of behavioral dynamics, and 2) too many features may introduce noise for MVA. To solve this problem, we are proposing the feature selection method after tensor reduction.

## Methods

### Filter and Embedded Method

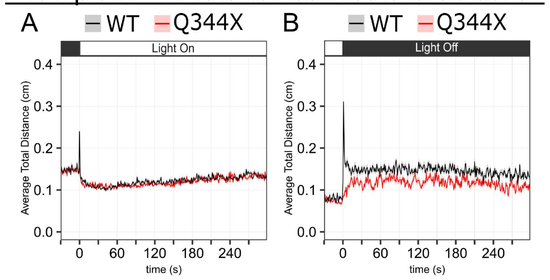
Intersection: Filter ∩ Embedded

Union: Filter ∪ Embedded

We aim to establish a feature selection workflow using Filter and Embedded methods for their low computational cost compared to the wrapper methods. For the filter method, we used the p-value t-test and fold change to select features. For the embedded method, we used the random forest to select features. We used two ways to integrate the feature selection results: intersection and union.

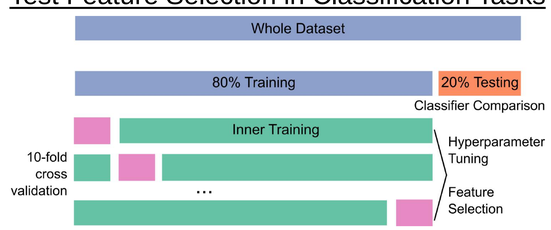
## Methods (cont.)

### Example VMR Data to Test Feature Selection




We tested the proposed feature selection methods in an example VMR using wild-type(WT) and a transgenic zebrafish line [*Tg(rho: Hsa.RH1_Q344X)* ] (Q344X) at 7 days post-fertilization. Q344X larvae showed a reduced VMR. Data were collected from 10 biological replicates with 24 larvae per genotype per replicate.

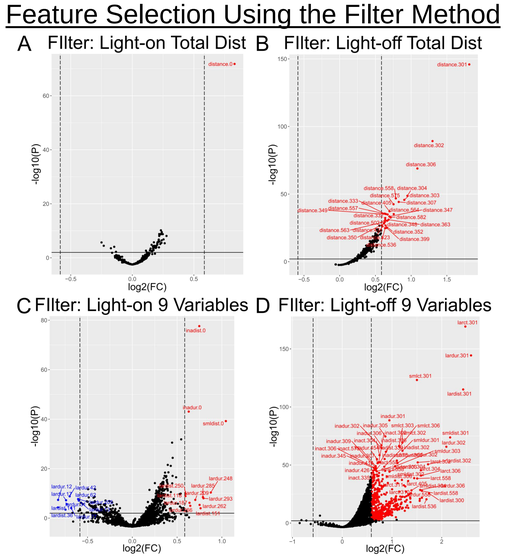
### Test Feature Selection in Classification Tasks




To evaluate if feature selection improved MVA after tensor reduction, we apply the selected features in a classification task to differentiate WT from Q344X using their VMR data. We first split the data into two subsets: the training and testing sets. In the training set, we applied 10-fold cross-validation (CV) to tune hyperparameters and compared classification accuracy before and after feature selection. After selecting the best hyperparameters and features, we built the classifiers using the training set and evaluated their performances in the testing set. We applied the following classifiers: Support Vector Machine (SVM), K-Nearest Neighbor (KNN) classifier, Naïve Bayes (NB), decision tree (DT), Random Forest (RF), and extreme gradient boosting (XGB).

## Results

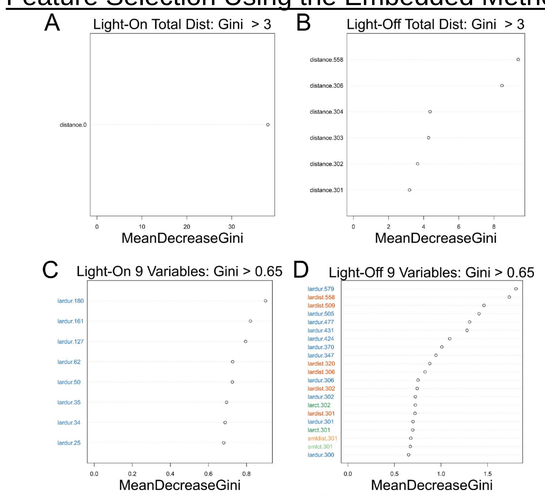
### Feature Selection Using the Filter Method




## Results (cont.)

For the Filter method, we used p-value from the t-test (y-axis) and fold change (x-axis) to select features where WTs differed from Q344X mutants. The cut-offs are 0.01 for P and 1.5 for FC. In the figure, both P and FC were log-transformed. Red dots represent features where WTs significantly differed from Q344X mutants and vice versa in blue. Black dots represent non-significant features.

### Feature Selection Using the Embedded Method




For the Embedded method, we used RF to select features that best explain the difference between WT and Q344X. In both tensor reduction methods, we used 3 and 0.65 for the cut-off for the mean Gini decrease, respectively. The cut-offs were based on where substantial value decrease, i.e., "elbow," occurred.

### CV Accuracy Using Total Distance: Before v.s. After Feature Selection

| | Classifier | Without | Intersect | Union |
|---|---|---|---|---|
| Light-On | SVM | **0.707291** | 0.699722 | 0.699722 |
| | NB | 0.611618 | **0.679663** | **0.679663** |
| | KNN | 0.695411 | **0.742521** | **0.742521** |
| | DT | **0.709873** | 0.708404 | 0.708404 |
| | RF | **0.719101** | 0.624108 | 0.624108 |
| | XGB | **0.735601** | 0.707827 | 0.707827 |
| Light-Off | SVM | **0.782623** | 0.778484 | 0.764804 |
| | NB | 0.70428 | **0.733459** | 0.679663 |
| | KNN | 0.599047 | **0.764763** | 0.749995 |
| | DT | 0.684832 | **0.765263** | 0.734703 |
| | RF | 0.80144 | 0.773464 | **0.806524** |
| | XGB | 0.706868 | **0.778653** | 0.75875 |

We first tested if feature selection increases classification accuracy using total distance as the behavioral variable. The accuracy of most classifiers was higher with feature selection. The largest number in each row is bolded.

### CV Accuracy Using 9 Variables: Before v.s. After Feature Selection

| | Classifier | Without | Intersect | Union |
|---|---|---|---|---|
| Light-On | SVM | 0.743253 | 0.616397 | **0.848604** |
| | NB | 0.749526 | 0.834126 | **0.85337** |
| | KNN | 0.637015 | **1** | 0.859509 |
| | DT | 0.901529 | **1** | 0.961413 |
| | RF | 0.991751 | **1** | **1** |
| | XGB | 0.88892 | **0.967635** | 0.966969 |
| Light-Off | SVM | 0.711501 | **0.805535** | 0.680577 |
| | NB | 0.693716 | 0.73407 | **0.763471** |
| | KNN | 0.433892 | **0.802662** | 0.493483 |
| | DT | 0.862784 | **0.89861** | 0.8855 |
| | RF | 0.914295 | **0.951092** | 0.931115 |
| | XGB | 0.841292 | **0.881862** | 0.878291 |

## Results (cont.)

We then tested if feature selection increases classification accuracy using all 9 behavioral variables. The accuracies of most classifiers were higher with feature selection. The largest number in each row is bolded.

### CV Accuracy after Feature Selection: 9 Variables v.s. Total Distance

| | Classifier | Distance only Best | 9 Variables Best |
|---|---|---|---|
| Light-On | SVM | 0.707291 | **0.848604** |
| | NB | 0.69663 | **0.85337** |
| | KNN | 0.742521 | **1** |
| | DT | 0.709873 | **1** |
| | RF | 0.719101 | **1** |
| | XGB | 0.735601 | **0.967635** |
| Light-Off | SVM | 0.782623 | **0.805535** |
| | NB | 0.733459 | **0.763471** |
| | KNN | 0.764763 | **0.802662** |
| | DT | 0.765263 | **0.89861** |
| | RF | 0.806524 | **0.951092** |
| | XGB | 0.778653 | **0.881862** |

We found that the classification accuracies were higher when using 9 variables with feature selection. The largest number in each row is bolded.

### Performance of Classifiers in Testing Set

| | | Acc. | Sens. | Spec. | Prec. | Kappa | AUROC |
|---|---|---|---|---|---|---|---|
| Light-On | SVM | 0.6591 | 0.6818 | 0.6364 | 0.6522 | 0.3182 | 0.6591 |
| | NB | 0.8068 | 0.9318 | 0.6818 | 0.7455 | 0.6136 | 0.8068 |
| | KNN | **1** | **1** | **1** | **1** | **1** | **1** |
| | DT | **1** | **1** | **1** | **1** | **1** | **1** |
| | RF | **1** | **1** | **1** | **1** | **1** | **1** |
| | XGB | **1** | **1** | **1** | **1** | **1** | **1** |
| Light-Off | SVM | 0.7955 | 0.7727 | 0.8182 | 0.8095 | 0.5909 | 0.7955 |
| | NB | 0.625 | 0.7045 | 0.5455 | 0.6078 | 0.25 | 0.625 |
| | KNN | 0.7614 | 0.75 | 0.7727 | 0.7674 | 0.5227 | 0.7614 |
| | DT | 0.8636 | 0.8182 | 0.9091 | 0.9 | 0.7271 | 0.8636 |
| | RF | **0.9205** | **0.8636** | **0.9773** | **0.9744** | **0.8409** | **0.9205** |
| | XGB | 0.8409 | 0.8409 | 0.8409 | 0.8409 | 0.6818 | 0.8409 |

After feature selection and hyperparameter tuning in CV, we finally built all classifiers using the training set and tested the performance of all classifiers on the testing set. Random Forest (RF) showed the highest performance. The largest number in each row is bolded.

## Conclusions

- We built a feature selection workflow to improve multivariate analysis after tensor reduction, facilitating the downstream research in neural processing underlying the dynamic behaviors.

- In the VMR analysis, we found that concatenating behavioral variables achieves higher classification performance than constructing one behavior variable when using feature selection.

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